Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and spending patterns.

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Dataset at a Glance

3,900	18	$59.76	3.75
Purchases	Features	Avg. Spend	Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings, Shipping

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Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing Review Ratings with category median

03

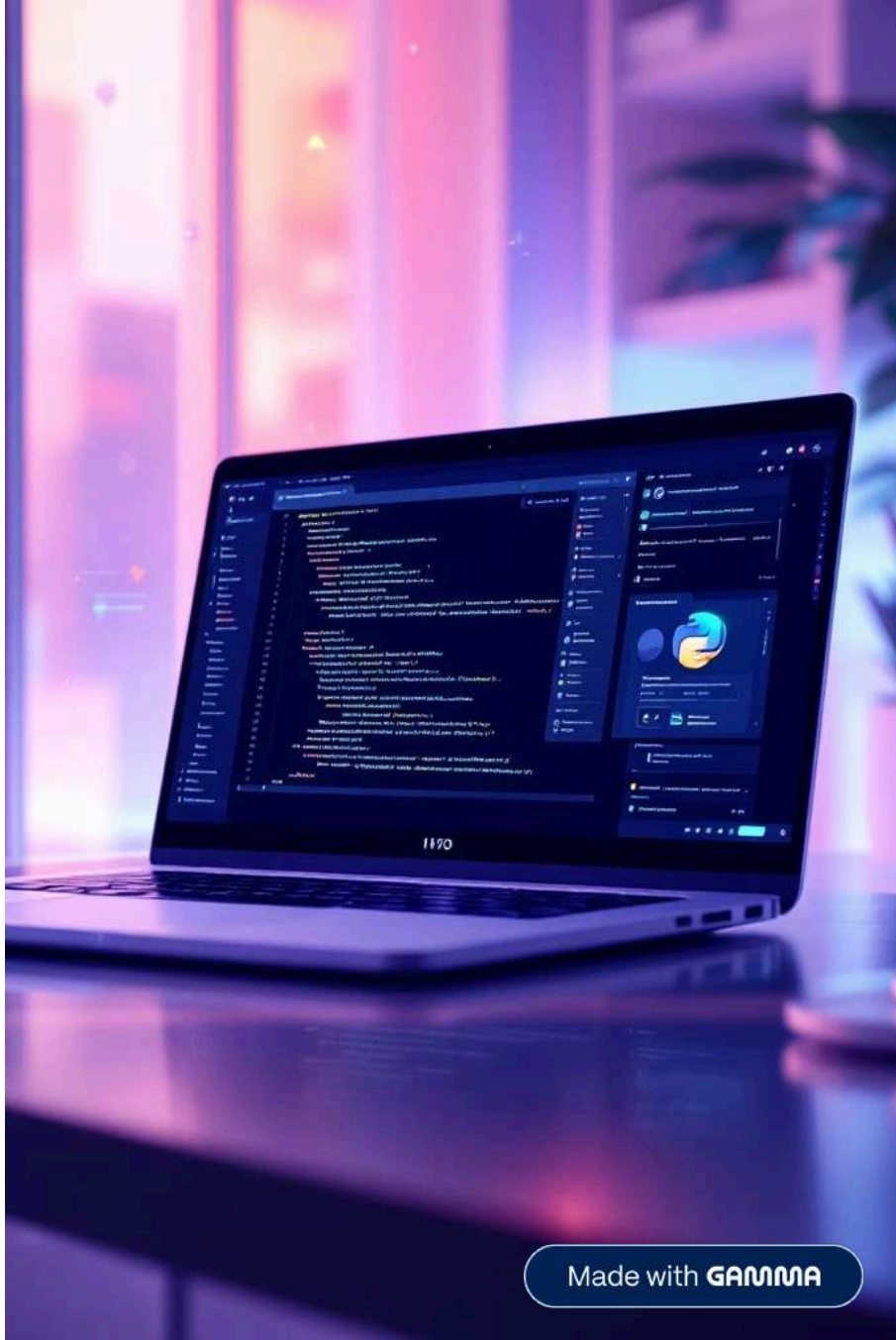
Feature Engineering

Created **age_group** and **purchase_frequency_days** columns

04

Load to PostgreSQL

Cleaned DataFrame exported to database for SQL analysis



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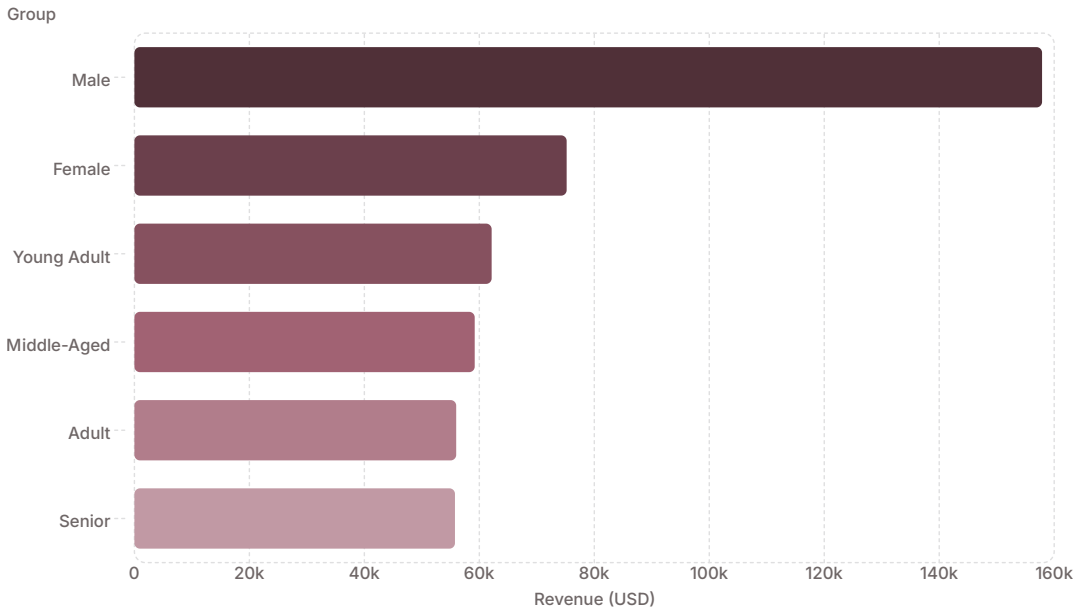
Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults lead revenue by age group
- Express shipping users spend slightly more ($60.48 vs. $58.46 Standard)
- 839 discount users still spent **above average**

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Top-Rated Products

Highest Avg. Review Ratings

Product	Avg. Rating
Gloves	3.86 ⭐
Sandals	3.84 ⭐
Boots	3.82 ⭐
Hat	3.80 ⭐
Skirt	3.78 ⭐

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Top Products by Category

👗 Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

💍 Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

👟 Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

🧥 Outerwear

- Jacket — 163
- Coat — 161

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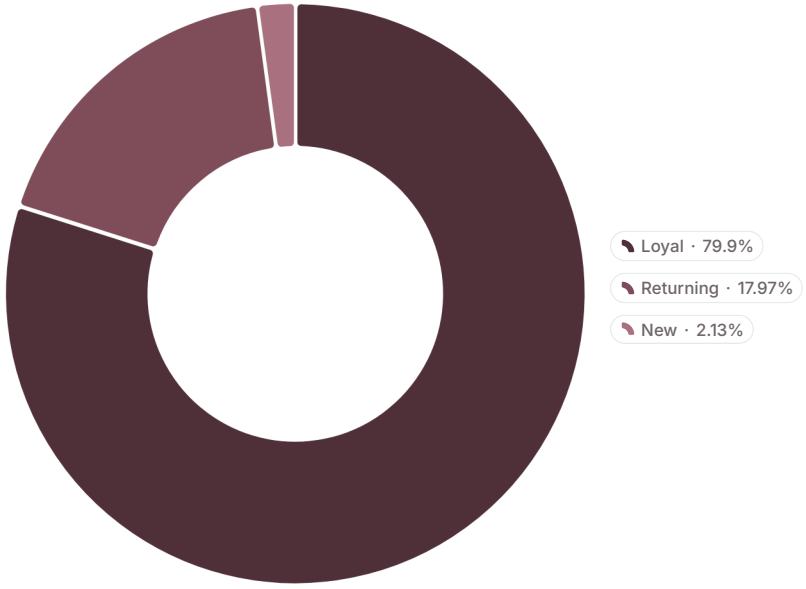
Customer Segmentation



Segment Breakdown

- **Loyal** — 3,116 customers (80%)
- **Returning** — 701 customers (18%)
- **New** — 83 customers (2%)

Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 non-subscribed

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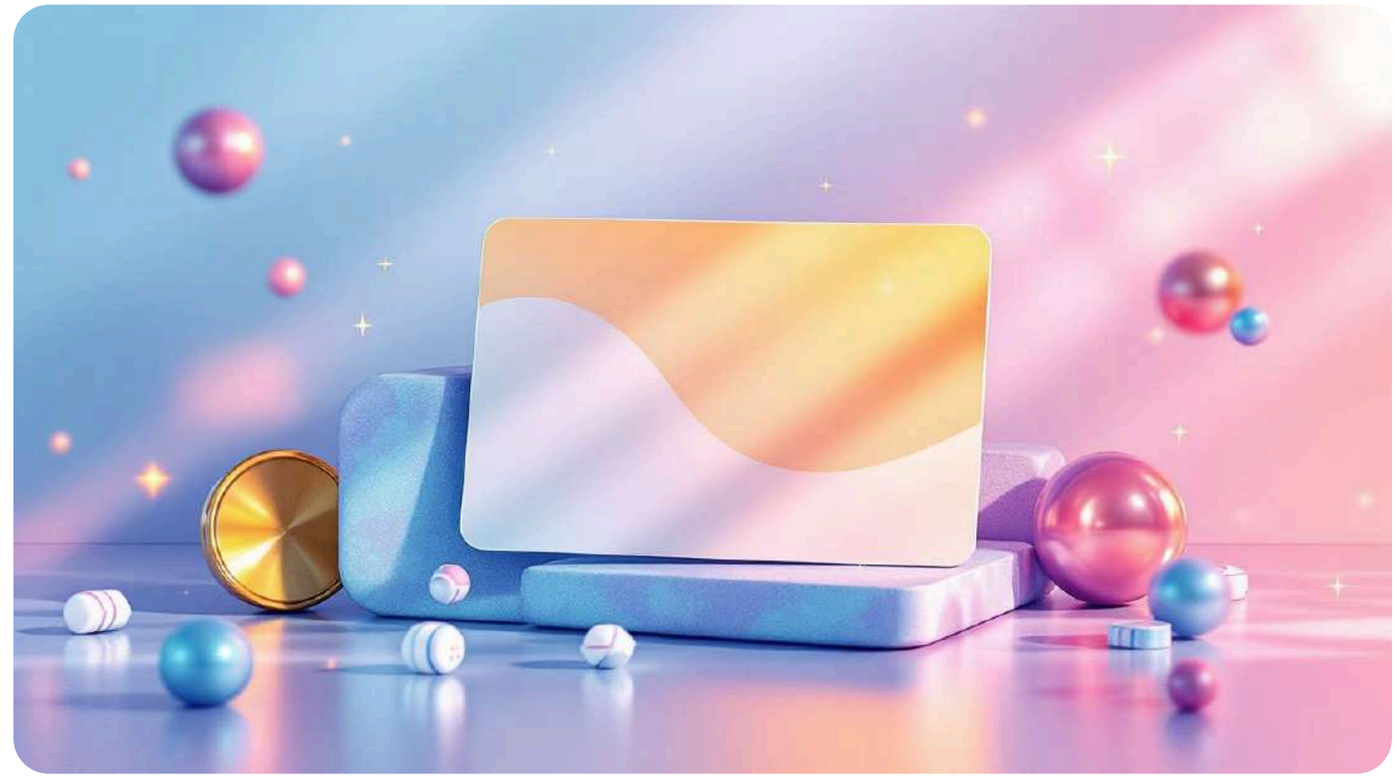
Subscription & Discount Behavior

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Most Discount-Dependent Products

Item	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%



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Power BI Dashboard

27% Subscribers

Donut chart: 73% No, 27% Yes

Revenue by Category

Clothing leads, followed by Accessories

Revenue by Age

Adults & Middle-Aged top contributors

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Business Recommendations

→ **Boost Subscriptions**

Promote exclusive subscriber benefits

→ **Loyalty Programs**

Reward repeat buyers to grow the Loyal segment

→ **Review Discount Policy**

Balance sales boosts with margin control

→ **Targeted Marketing**

Focus on high-revenue age groups & express-shipping users

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